Individual Project: Build a SLAM Application

Ling Wang

Mechanical Robotics Engineering, Southern Illinois University Edwardsville

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Professor Mingshao Zhang

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By reading the example webpage, it seems to me that the implementation steps of Visual-SLAM are mainly divided into the following steps:

1. Extract ORB features from two frames and match them. Use triangulation method to calculate the initial 3D map points and the relative pose of the camera.

2. Take the initial frame as the keyframe and store the relevant 3D map points. Store the camera pose at the same time.

3. Build Bag-of-Features data for loop detection.

4. Optimize the initial map and camera pose through bundleAdjustment. Visualize the initial 3D reconstruction results.

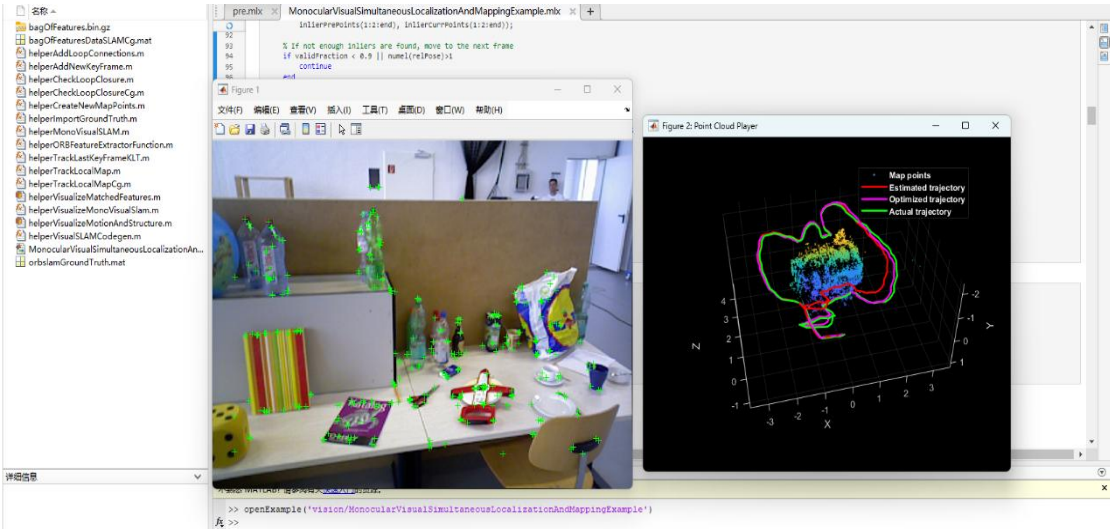
5. For each new frame, extract features and match them with the features of the previous keyframe.

6. Perform local mapping for each keyframe.

7. Use the current keyframe processed by the local mapping process and try to detect and close the loop.

8. Finally, perform similar pose graph optimization on the basic graph to correct the drift of the camera pose.

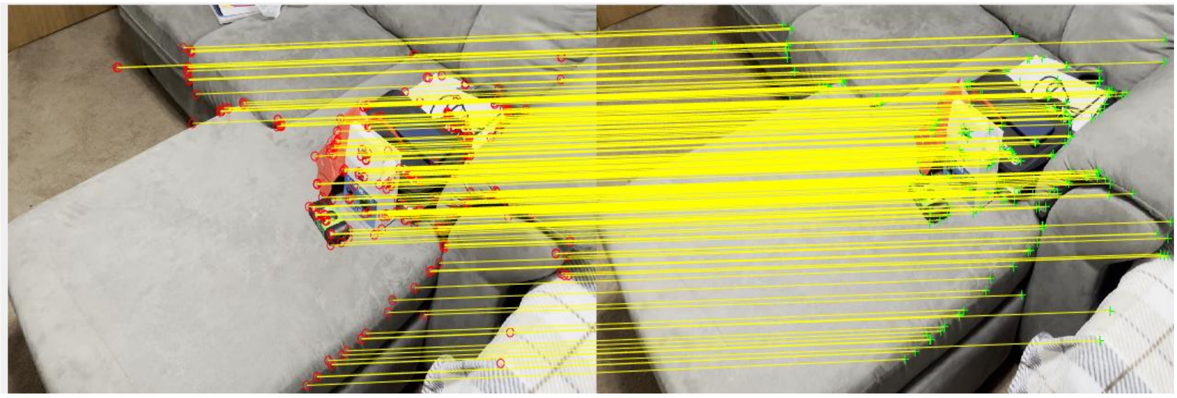
9. Use MATLAB Coder to generate multithreaded C/C++ code from the algorithm for deployment to non-PC hardware or running as a ROS node.

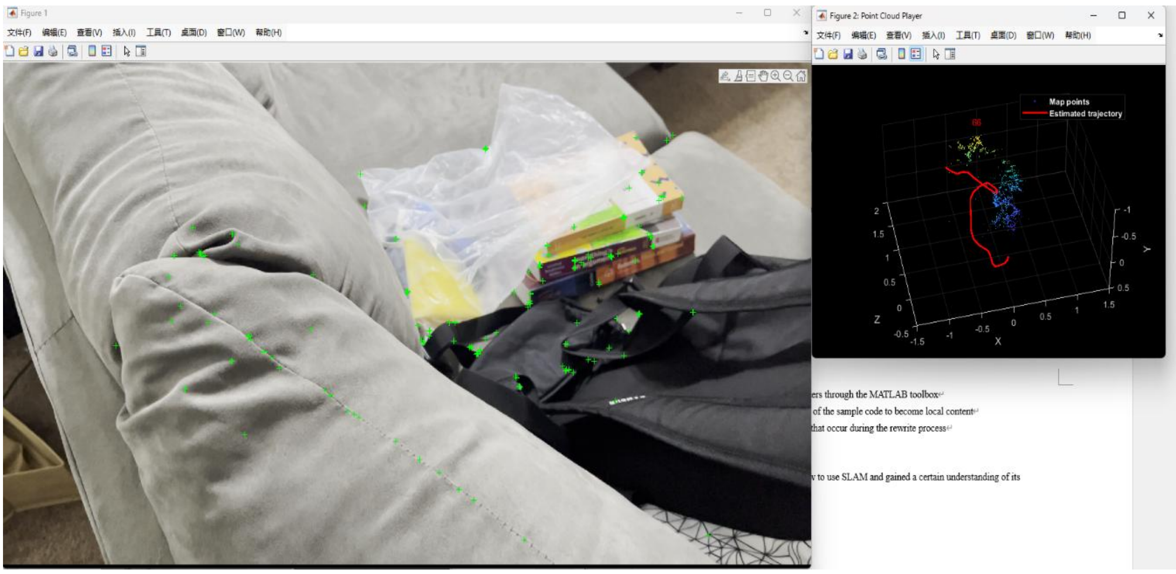


How to implement a visual-SLAM application

1. Use your mobile phone to record a video.
2. Get the camera parameters through the MATLAB toolbox
3. Rewrite the calling part of the sample code to become local content
4. Dealing with problems that occur during the rewrite process
5. Deal with problems that occur during code operation

In this project, I learned how to use SLAM and gained a certain understanding of its workflow. In this project, I learned how to use SLAM and had a certain understanding of its workflow. I modified the webpage download file in the example to load the video locally, and used VideoReader to read the video image frame by frame, and finally realized the display of the complete 3D camera and sparse path reconstruction points.





Questions about this project

In this project, I spent a long time dealing with bugs in the code, including how to use local videos to carry out this project, bagOfFeaturesDBoW error, variable 'monoVisualSLAM' cannot be recognized and Incorrect number or types of inputs or outputs for function codegen. Most of the problems were solved after a period of exploration, but the last problem still troubled me for a long time, and I was still struggling to debug it before the deadline.

I have heard of Github for a long time but have no idea about its specific use. So as it was my first time to use it, I spent some time searching for how to create libraries and import files.

Regarding this project, I was very happy when I first came across the content, because I am very interested in the technology in this area. I would also read related articles and watch videos when browsing the web. But I didn't expect it would be so difficult to implement it myself. I spent a lot of time but the code still had a stubborn problem that I couldn't solve, which made me feel frustrated. Of course, this also makes me respect technology more.

In short, thanks for this opportunity to allow me to personally experience how to use this algorithm to draw unknown environments.